Abdullah Naveed

Ph.D. Candidate in Computer Engineering | University of Florida

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github.com/Abdullahnaveed102 | Google Scholar

Research Interests

Dependable AI and high-performance computing; LLM reliability; compiler and runtime resilience; scientific data compression; CPU, GPU, and FPGA acceleration; hardware–software co-design.

Education

University of Florida **Aug 2025 – Expected May 2027**
Ph.D. Candidate in Computer Engineering *Gainesville, FL*
GPA: 3.93/4.00 | Advisor: Prof. Guanpeng Li | Dependable Systems Lab

University of Iowa **Aug 2023 – Aug 2025**
Master of Computer Science *Iowa City, IA*
GPA: 3.99/4.00

Lahore University of Management Sciences **Aug 2019 – May 2023**
B.S. in Computer Science *Lahore, Pakistan*
GPA: 3.61/4.00

Research Experience

University of Florida **Aug 2025 – Present**
Graduate Research Assistant, Dependable Systems Lab *Gainesville, FL*

- Develop methods for dependable AI and scientific computing, focusing on LLM resilience and efficient error detection. Advised by Prof. Guanpeng Li.
- Developed **TRACE**, a runtime silent-error detector achieving **95.3% average detection coverage** across 15 LLMs and 3 datasets with **0.56% runtime overhead**; paper accepted at SC26.
- Characterize the effects of transient hardware faults on LLM inference, including models adapted to new tasks, and evaluate reliability and protection costs.

Argonne National Laboratory **Jun 2025 – Present**
Visiting Graduate Researcher *Remote*

- Develop scientific data compression and data systems for efficient scientific computing, in collaboration with Dr. Sheng Di and Prof. Guanpeng Li.
- Design and evaluate CPU, GPU, and FPGA acceleration through parallel computing and hardware–software co-design, with a focus on compression throughput, data fidelity, and data movement.

University of Iowa **Aug 2023 – Aug 2025**
Graduate Research Assistant *Iowa City, IA*

- Analyzed hardware-fault effects in scientific and HPC applications and evaluated compiler and runtime approaches to detecting silent data corruption.
- Measured trade-offs between detection effectiveness and execution cost across scientific workloads using fault injection and experimental evaluation.

Undergraduate Research Collaboration **Aug 2022 – May 2023**
LUMS · University of Arizona · SRI International

- Evaluated container debloating tools for attack-surface reduction, vulnerability mitigation, image size, and execution correctness; co-authored *Evaluating Container Debloaters* (IEEE SecDev 2023).
- Collaborated with Prof. Fareed Zaffar, Prof. Sazzadur Rahaman, and Dr. Ashish Gehani across LUMS, the University of Arizona, and SRI International (formerly Stanford Research Institute).

Publications

1. **Abdullah Naveed**, Mahina Sheikh, and Guanpeng Li.
   Low-Overhead Silent Error Detection for Large Language Model Inference at Scale.
   International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26), 2026.
   Accepted; to appear.

2. Darren Ng, **Abdullah Naveed**, Duo Zhang, Guanpeng Li, Sheng Di, and Xiaoyi Lu.
   Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs.
   International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26), 2026.
   Accepted; to appear.

3. Muhammad Hassan, Talha Tahir, Muhammad Farrukh, **Abdullah Naveed**, Anas Naeem, Fareed Zaffar, Fahad Shaon, Ashish Gehani, and Sazzadur Rahaman.
   Evaluating Container Debloaters.
   2023 IEEE Secure Development Conference (SecDev), pp. 88–98, 2023.

Teaching Experience

University of Iowa **Aug – Dec 2023**
Teaching Assistant, Database Systems *Iowa City, IA*
- Mentored more than 30 students through MySQL projects, relational schema design, query debugging, and implementation decisions during labs and office hours.

Lahore University of Management Sciences **Aug – Dec 2022**
Teaching Assistant, Computational Problem Solving *Lahore, Pakistan*
- Supported more than 100 students through tutorials, programming exercises, assignment feedback, and assessments.

Professional Service

- Co-reviewed conference submissions with my advisor for SC, IPDPS, DSN, ICS, AAAI, and HPDC.
- Contribute to ProdeBench through implementation and experimental evaluation.

Selected Technical Projects

DebloatBench / BloatProfiler
Developed a Python and Docker benchmarking framework for container debloaters, evaluating attack surface, vulnerability mitigation, image size, and execution correctness. Published at IEEE SecDev 2023.

Multimodal Fake News Classification
Developed and benchmarked text–image classifiers using BERT, CNN vision models, and multimodal fusion; evaluated accuracy, precision, recall, and F1.

LUMS Mobile App
Developed a cross-platform student application with React Native, JavaScript, and Firebase, integrating deadline management, GPA tracking, communication, and real-time data.

Technical Skills

Programming:	Python, C, C++, Scala, Bash, SQL, JavaScript, Git
Machine Learning / Data:	PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
Systems / HPC:	LLVM, Linux, Docker, MPI, OpenMP, Slurm, fault injection, parallel computing
Hardware:	Chisel, Verilog/RTL, RISC-V, Chipyard, Verilator, Vivado
Software Development:	React, React Native, Firebase, HTML/CSS